Gearbox fault prediction using smartphone voice recording

A visual companion to the Jupyter implementation | Prepared 13 October 2026

Purpose

Understand gear-tooth and bearing faults and how to record gearbox sound consistently with a smartphone. This illustrated guide explains the fault categories, test rig and proposed recording layout for the minor project.



Figure 1. Gear train and supporting shafts. This internal view illustrates the gearbox arrangement; it is not an operating recording session.

Three labels used by the notebook

NORMAL Reference operation with no known introduced tooth or bearing fault.

TOOTH FAULT Damage to a gear tooth or its working surface; examples include breakage, pitting and cracking.

BEARING FAULT Damage to a bearing raceway or rolling element; several physical locations share this one label.

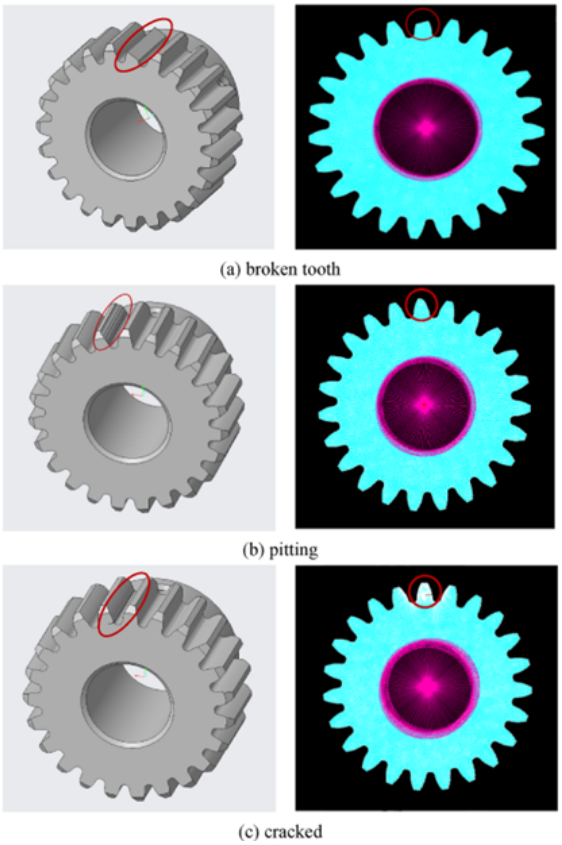
Guide map: gear-tooth damage (p. 2), bearing damage (p. 3), test rig (p. 4), recording layout (p. 5), and model scope (p. 6).

MINOR PROJECT
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01 Gear-tooth faults

What changes in the tooth, and how the fault types differ

Gears transmit torque through repeated tooth contact. Damage can affect the tooth body, its root, or its contact surface. The examples below describe physical fault types; the notebook groups them under **tooth_fault**.



Broken or chipped tooth

Part or all of a tooth is missing. Fracture may follow overload or progressive fatigue; the missing material changes contact and load sharing.

Pitting and spalling

Contact fatigue produces small pits on a working flank. Larger patches of material loss are often described as spalling.

Tooth or root cracking

A crack can grow under repeated loading, including bending near the root. A cracked tooth may remain present before it fractures.

Figure 2. Broken-tooth, pitting and cracked-tooth examples. These illustrate fault categories, not dimensions or severity measurements.

Other surface damage to recognize

Scuffing / scoring. Lubricant-film breakdown can produce adhesive damage and marked tooth surfaces.

Abrasive wear. Particles in the contact or lubricant can remove material and alter the tooth profile.

Interpreting sound

Changes in contact can affect sound and vibration, but an audible change alone does not identify the damage mechanism. Compare the same speed and load, and confirm labels from component inspection. The current model does not predict fault severity.

02 Bearing faults

Separate the damaged location from the damage mechanism

Bearings support rotating shafts. The inner ring, outer ring and rolling elements have different roles, but each can contain a localized defect. The notebook uses one combined **bearing_fault** label for these locations.

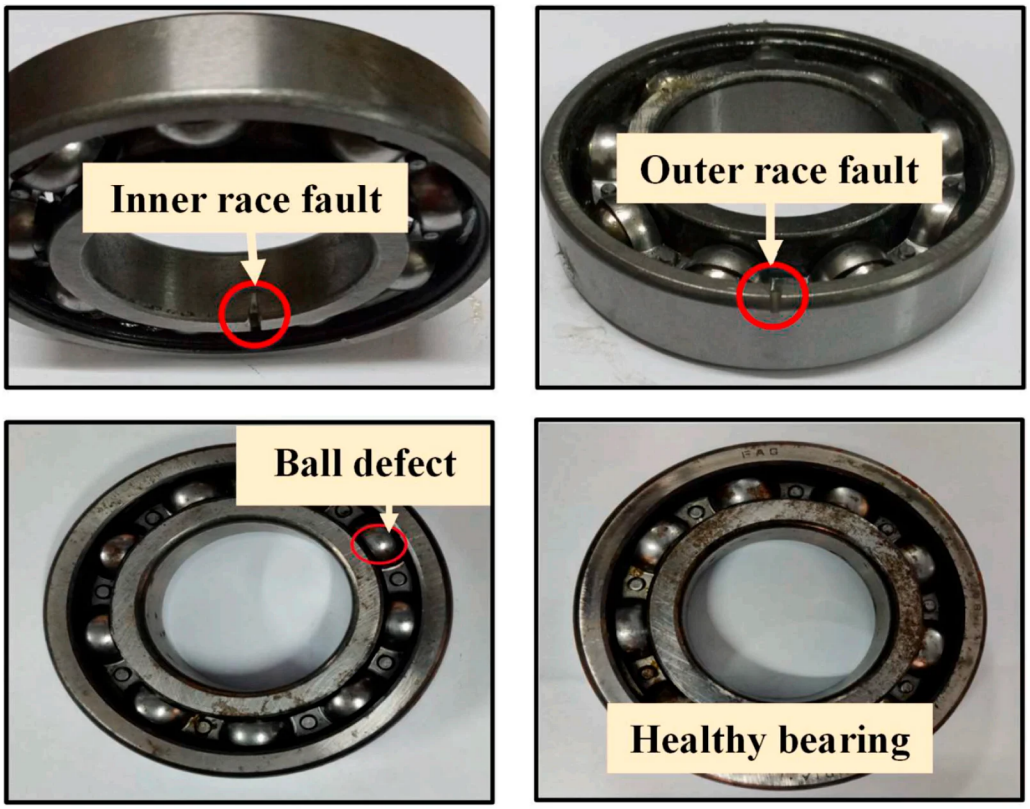


Figure 3. Inner-race, outer-race and ball-defect examples alongside a healthy-bearing illustration. Naturally developed damage may look different from these localized defects.

Inner race. The ring mounted on the shaft; defects occur along its rolling contact track.

Outer race. The ring supported by the housing; a defect affects the track contacted by passing rolling elements.

Rolling element. A ball or roller may have a damaged surface that disrupts smooth rolling.

Common mechanisms

Flaking / spalling: rolling fatigue removes small pieces from a raceway or rolling element. Poor lubrication, misalignment, excessive load and contamination can contribute. **Indentation:** debris pressed through a contact can damage the surface. **Wear:** contamination and inadequate lubrication can change surface condition and promote further distress.

A location label is not a root-cause diagnosis. For example, an outer-race fault may involve different damage mechanisms. Inspection remains necessary.

03 Gearbox test rig

Gearbox test-bench components and their roles



Figure 4. Test-bench photograph showing the drive and gearbox arrangement. The recommended microphone position is drawn separately on page 5.

Read the photograph from left to right

Drive motor and controller. The motor is visible at lower left, with its drive/control unit above it. Set the intended operating condition and record the measured RPM.

Coupling and guard. The yellow cover lies between the motor and gearbox. Keep rotating connections guarded during acquisition.

Gearbox housing. The central enclosure contains the gear train. Place the phone consistently near the same casing surface.

Output-side unit. A coupled unit is visible at the right. Its exact type and load calibration are not established by the image; document the actual laboratory hardware.

Additional recording equipment

Use a smartphone with WAV recording, a fixed stand, an RPM measurement method, a record of load/settings, and a computer for the notebook. Use an approved laboratory rig and isolate power before changing or inspecting components.

04 Smartphone recording layout

Proposed recording arrangement; schematic, not to scale

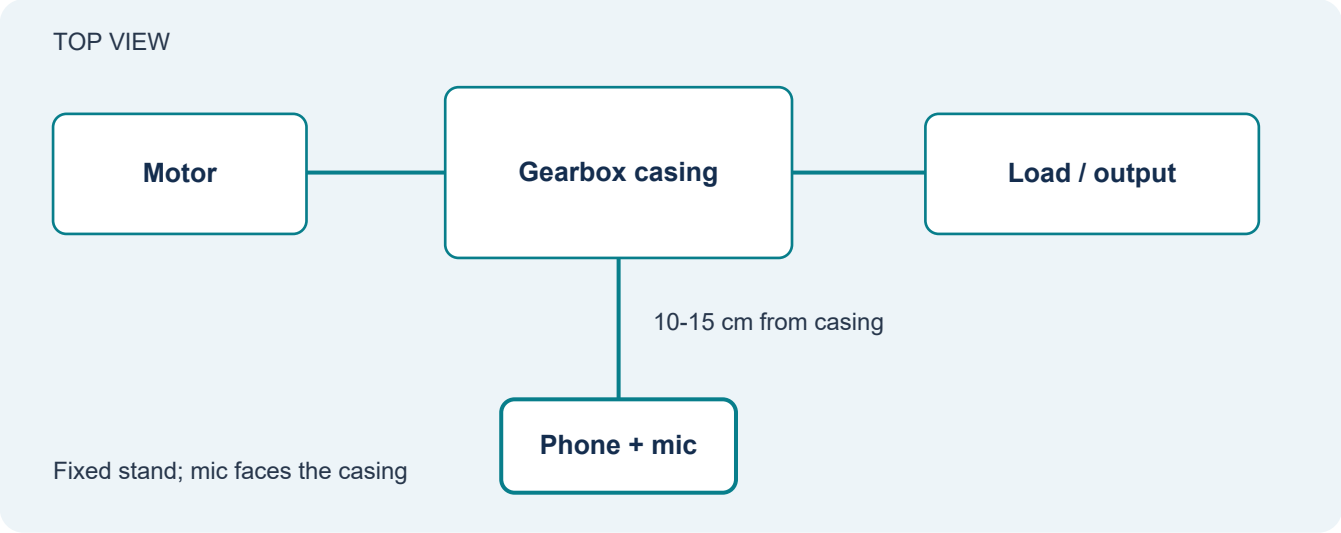


Figure 5. Original vector schematic. Position the microphone near casing height with fixed orientation and separation, while keeping it clear of moving parts. The diagram represents the proposed setup, not a measured installation.

Recording procedure

1. Confirm the condition. Inspect and document the component state before recording. Use the known condition as the label.

2. Fix the geometry. Use the same phone, stand position, casing surface and microphone orientation for every class.

3. Stabilize operation. Set the target speed and load, allow operation to settle, and log the actual RPM and settings.

4. Record and inspect. Capture WAV audio at 44.1 or 48 kHz if available. Check for clipping, interruptions and strong background noise.

5. Repeat independently. Collect separate sessions across normal, tooth-fault and bearing-fault conditions, including the proposed 300, 500 and 700 RPM settings.

Minimum notebook metadata: **path, label, group_id, rpm**. Keep all clips from one acquisition session in the same group. Also log phone model, load, distance, date and fault evidence in your experiment notes.

05 From physical faults to labels

How this guide connects to the notebook

The notebook resamples audio to 16 kHz, forms two-second windows, and extracts 13 MFCCs plus first and second derivatives. CNN-LSTM, RNN and GRU use the resulting sequences; SVM and Gaussian Naive Bayes use summary features.

What the model can and cannot say

Three outputs only. normal, **tooth_fault** and **bearing_fault**. It does not distinguish a tooth crack from pitting, or an inner-race fault from an outer-race fault.

Controlled comparison. Split independent recordings or sessions before segmenting them. Compare models on the same held-out groups, with matched speed/load documentation.

Experimental evidence. Current notebook examples use synthetic signals. Neither those scores nor these illustrations establish real diagnostic accuracy. Unknown faults, combined faults and severity estimation require additional labels and validation.

Project details

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Setup values are project proposals, not universal diagnostic requirements. This companion explains the experiment; the Jupyter notebook remains the implementation source.